

Advanced Statistics (DS2003)

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Course Instructor(s)

Dr. Muhammad Ahmad Raza

Sessional-I Exam

Total Time: 1 Hours

Total Marks: 60

Total Questions: 03

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Campus: Lahore

Dept: School of Computing

Excellent!

Menahil Amir
Student Name22L-7459 4B
Roll No SectionMenahil
Student Signature

Vetted by

Vetter Signature

CLO 1: CLO statement for question Q1&Q2

Learn about basic statistical methods

Q1: Question statement [15+15=30]
 Suppose that orders at a restaurant are IID random variables with mean $\mu = \$8$ and standard deviation $\sigma = \$2$. Answer the followings:

- a) Estimate the probability that the first 100 customers spend a total of between \$780 and \$820.

$$\mu = 8 \quad \sigma = 2$$

$$a) P(780 < T < 820)$$

$$\mu_T = n\mu = 100 \times 8 = 800$$

$$\sigma_T^2 = n\sigma^2 = 100 \times 4 = 400$$

$$\sigma = 20$$

$$Z = (X - \mu) / \sigma \sqrt{n}$$

$$\rightarrow P\left(\frac{780 - 800}{20} < Z < \frac{820 - 800}{20}\right)$$

$$= P(-1 < Z < 1) = P(Z < 1) - P(Z < -1)$$

from table $\Rightarrow 0.8413 - 0.1587$

$$P(780 < T < 820) = 0.6826$$

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b) After how many orders can we be 90% sure that the total spent by all customers is more than \$1000?

$$P(T > 1000) = 0.9$$

$$1 - P(T < 1000) = 0.9$$

$$P(T < 1000) = 0.1$$

$$\mu = 8$$

$$\sigma_T = 8n$$

$$\sigma = 2$$

$$\sigma_T = \sqrt{n} \times 2$$

$$Z = \frac{T - \mu_T}{\sigma_T} \quad \text{at } P = 0.1$$

$$Z = -1.28$$

$$-1.28 = \frac{1000 - 8n}{2\sqrt{n}}$$

$$(-2.56\sqrt{n})^2 = (1000 - 8n)^2$$

$$6.5536n = 1000000 + 64n^2 - 16000n$$

$$X = 128.63, 121.47$$

90% sure. Why not this value?

$$N = 129 \text{ customers}$$

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$$0.9$$

$$\frac{1000 - 8 \times 129}{\sqrt{4 \times 129}}$$

$$-1.4087$$

$$0.92$$

Q2: Question statement[10]

The PMF, mean and variance of a Poisson random variable are given as follows:

$$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$E(X) = \lambda; \quad \text{Var}(X) = \lambda$$

Now consider a population following the Poisson distribution with mean of 2. Suppose that a sample of size 16 is obtained from this population. Use the central limit theorem to estimate the probability that the sample mean is greater than 2.5.

$$\mu = E(X) = 2, \quad N = 16$$

$$E(X) = \lambda$$

$$2 = \lambda$$

$$\text{var}(X) = 2$$

$$\sigma(X) = \sqrt{2}$$

$$P(\bar{X} > 2.5) = 1 - P(\bar{X} < 2.5)$$

$$\mu_{\bar{X}} = \mu = 2$$

$$\sigma_{\bar{X}} = \sigma / \sqrt{n} = \sqrt{2} / \sqrt{16} = 0.35355$$

$$P(\bar{X} < 2.5) = P\left(Z < \frac{2.5 - 2}{0.35355}\right)$$

$$= P(Z < 1.414)$$

$$\text{from table } P(Z < 1.414) = 0.9207$$

$$P(\bar{X} < 2.5) = 1 - 0.9207$$

$$= 0.0793$$

CLO 2: CLO statement for question Q3

Understand the reasons for developing confidence intervals and performing hypothesis tests

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Q3: Question statement

[10+10=20]

- a) An electrical firm manufactures light bulbs. The life of each bulb is approximately normally distributed with a standard deviation of 40 hours. If a sample of 30 bulbs has an average life (sample mean) of 780 hours, find a 96% confidence interval for the population mean of all bulbs produced by this firm.

$\sigma = 40$, $n = 30$, $\bar{x} = 780$.
96% confidence interval.

$$\alpha = 1 - 0.96 = 0.04$$

$$\alpha/2 = 0.02$$

value of $z_{\alpha/2} = \pm 2.05$

$$z_{\alpha/2} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\bar{x} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$780 - 2.05 \times \frac{40}{\sqrt{30}} < \mu < 780 + 2.05 \times \frac{40}{\sqrt{30}}$$

Confidence interval = $768.029 < \mu < 794.97$

- b) How large a sample is needed if we wish to be 96% confident that our sample mean will be within 10 hours of the true mean?

$$\text{error} = Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$10 = 2.05 \times \frac{40}{\sqrt{n}}$$

$$n = \left(\frac{2.05 \times 40}{10} \right)^2$$

$$n = 67.24 \rightarrow \underline{\underline{68}}$$

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